

BSIT MELEC 9: System Need Analysis

Module 4 Study Reviewer & Practice Examination

COURSE: BSIT - MELEC 9 (System Need Analysis)

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(Learning Sheet No. 4)

TOPIC: Requirements Gathering Techniques

COVERAGE: Prelim Examination

Key Competencies: Master the 4 primary elicitation techniques: Interviews, Observation, Questionnaires, and Document Analysis; strengths, weaknesses, and hybrid strategies.

Format: Notes + Practice Test + Answer Key

I. Concept and Critical Importance of Requirements Gathering

Requirements Gathering (Elicitation) is the systematic process of collecting, discovering, and refining information from stakeholders to determine exactly what a new software system must accomplish and how it should perform.

Why is it Considered the Most Critical Phase?

Requirements errors represent over 50% of all software project failures. Building the *wrong system* perfectly is completely useless. Requirements gathering ensures developers solve the right business problems, prevent feature bloat (gold plating), and avoid catastrophic post-release redesigns.

II. The Four (4) Primary Elicitation Techniques

Technique	Core Definition & Subtypes	Key Advantages	Key Limitations
1. Interview	Direct conversation (face-to-face, video, phone) between analyst and stakeholder. • Structured: Fixed, standardized questions. • Unstructured: Open, conversational dialogue. • Semi-Structured: Core questions + flexible probes (Most common).	• Rich qualitative depth. • Immediate clarification. • Builds rapport & trust. • Uncovers hidden political/operational concerns.	• Time-consuming. • Expensive to conduct. • Subject to interviewer/interviewee bias. • Reaches few people.
2. Observation	Watching users execute daily workflows in their actual work environment. • Passive: Analyst observes silently without interruption. • Active: Analyst asks clarifying questions during the workflow.	• Shows actual reality vs what users claim. • Uncovers undocumented workarounds. • Independent of user memory or bias.	• Hawthorne Effect: Users alter habits when watched. • Time-intensive. • Cannot capture rare edge-cases. • Confidentiality limits.
3. Questionnaire (Survey)	Standardized written questions distributed to a large respondent pool. • Closed-Ended: Multiple-choice, Likert rating scales. • Open-Ended: Freeform essay answers.	• Highly cost-effective. • Rapid data collection. • Reaches hundreds/thousands of distributed users. • Easy quantitative analysis.	• Low response rates (10–30%). • Cannot clarify confusing questions. • Inflexible; lacks qualitative depth. • Risk of dishonest answers.
4. Document Analysis	Inspecting existing organizational artifacts: paper forms, invoices, receipts, policy manuals, audit reports, database schemas.	• Objective historical record. • Very low financial cost. • Zero interruption of employees. • Discovers exact data fields.	• Documents may be outdated or obsolete. • Cannot explain <i>why</i> a process exists. • Unofficial workarounds are not documented.

EXAM TIP / CRITICAL CONCEPT: PASSIVE VS. ACTIVE OBSERVATION & THE HAWTHORNE EFFECT

Exam questions frequently contrast **Passive Observation** (silent, unobtrusive watching) with **Active Observation** (asking 'Why did you write that down?' while the user works). Always remember the *Hawthorne Effect*: people often act more diligently, strictly follow rules, or change their normal behavior when they know an analyst is watching them!

III. Strategic Technique Selection: When to Use Which?

Project Scenario / Context	Recommended Technique	Professional Justification
You need to identify the exact data fields required on an official transcript.	Document Analysis	Reviewing the current official paper transcript reveals all mandatory fields, layouts, and signatures without interrupting staff.
You need feedback from 3,500 college students across 3 campus branches.	Online Questionnaire (Survey)	A digital survey collects large-scale statistical feedback rapidly and cheaply from geographically dispersed users.
A department head has complex, conflicting opinions on system workflow.	Semi-Structured Interview	Allows one-on-one deep exploration, probing into motivations, resolving doubts, and building executive rapport.
A cashier claims their daily process is simple, but queues stretch out the door.	Passive / Active Observation	Watching the actual cashier desk reveals hidden bottlenecks (e.g. printer jamming, double data-entry) that the user overlooked.

The Professional Multi-Method Sequence:

Top system analysts never rely on a single technique. The industry-standard sequence is:

- Step 1: Document Analysis** — Study existing forms, reports, and manuals to gain baseline domain knowledge.
- Step 2: Observation** — Watch staff perform daily work to see actual workflows and compare against written documents.
- Step 3: Interviews** — Meet key managers, registrars, and cashiers to clarify discrepancies and probe deep requirements.
- Step 4: Questionnaires** — Survey the general student or faculty body to statistically validate findings across a large population.

MODULE 4: SAMPLE EXAMINATION

Test your mastery of Module 4. Complete all parts before checking the Answer Key.

PART I: MULTIPLE CHOICE (10 Items)

Select the best answer for each question.

- 1. Which requirements elicitation technique is most effective when an analyst needs to gather opinions from 5,000 enrolled university students?**
 - A. In-depth face-to-face interviews
 - B. Online questionnaire / survey
 - C. Active observation of every student
 - D. Reviewing university financial ledgers
- 2. An analyst sits silently beside a billing clerk for two hours without asking questions or speaking. This is an example of:**
 - A. Active Observation
 - B. Passive Observation
 - C. Structured Interview
 - D. Document Sampling
- 3. What major psychological phenomenon describes users altering their regular work behavior because they know an analyst is watching them?**
 - A. Placebo Effect
 - B. Hawthorne Effect
 - C. Halo Effect
 - D. Murphy's Law
- 4. Which interview type utilizes a fixed, pre-written list of questions asked in identical sequence to all interviewees to facilitate direct comparison?**
 - A. Unstructured Interview
 - B. Structured Interview
 - C. Semi-structured Interview
 - D. Focus Group Discussion
- 5. If an analyst wants to uncover the exact fields printed on a school's official billing invoice without interrupting employees, which technique is ideal?**
 - A. Document Analysis
 - B. Student Survey
 - C. Active Observation
 - D. Brainstorming Session
- 6. What is the single most commonly used interview style in modern systems analysis because it balances structure with conversational flexibility?**
 - A. Completely Unstructured
 - B. Semi-Structured
 - C. Strictly Scripted
 - D. Cross-Examination
- 7. Which of the following is a recognized disadvantage of distributing questionnaires?**
 - A. Extremely high cost per respondent
 - B. Cannot clarify questions that respondents misunderstand
 - C. Requires the analyst to travel to every respondent
 - D. Impossible to analyze statistically
- 8. During requirements gathering, an analyst asks a cashier: 'Why do you re-write this student number on paper before typing it into the computer?' Which technique is this?**
 - A. Passive Observation
 - B. Active Observation

- C. Document Archival
- D. Closed Survey

9. Why is reviewing organizational documents alone insufficient for gathering all system requirements?

- A. Documents are illegal to read without a court warrant.
- B. Documents often describe idealized formal policies rather than actual day-to-day workarounds.
- C. Documents cannot contain numbers.
- D. Modern systems never use paper records.

10. In requirements gathering, questions where respondents must choose from predefined choices (e.g. Yes/No or Rating 1-5) are called:

- A. Open-ended questions
- B. Closed-ended questions
- C. Rhetorical questions
- D. Exploratory questions

PART II: TECHNIQUE MATCHING (5 Items)

Match each scenario with the most suitable technique: (A) Interview, (B) Observation, (C) Questionnaire, or (D) Document Analysis.

- 11. Eliciting deeply nuanced, strategic project expectations and institutional goals directly from the University President: _____
- 12. Discovering that warehouse staff regularly bypass an error message by unplugging the barcode scanner: _____
- 13. Extracting statutory tax calculation formulas and deduction rules from the national employee benefits manual: _____
- 14. Measuring overall customer satisfaction with a banking mobile app across 10,000 nationwide mobile app users: _____
- 15. Reconciling a situation where two senior managers provide contradictory descriptions of the approval chain: _____

PART III: TRUE OR FALSE (5 Items)

Write TRUE if the statement is correct; write FALSE if the statement is incorrect.

- 16. Closed-ended questions in questionnaires provide rich qualitative descriptions but are nearly impossible to analyze statistically.
- 17. The primary reason for combining multiple requirements gathering techniques is that each technique has unique blind spots and limitations.
- 18. In active observation, the system analyst never speaks to or interrupts the employee while they work.
- 19. Document analysis can reveal official business rules, but it often fails to uncover unofficial employee shortcuts.
- 20. Requirements gathering is only necessary if the software developers are completely unfamiliar with the programming language.

PART IV: SCENARIO APPLICATION & ESSAY (2 Items)

Read each practical scenario and answer thoroughly.

Question 21 (University Enrollment System Strategy):

You are appointed as the Lead System Analyst for a university developing a brand-new Online Enrollment System. **Which requirements gathering technique would you use FIRST?** Justify your choice by explaining how starting with this specific technique establishes a strong foundation and helps you effectively plan your subsequent elicitation activities.

Question 22 (Comparing Techniques):

Compare and contrast **Interviews** and **Questionnaires** in terms of: (a) Depth of information, (b) Cost and time requirements, (c) Flexibility, and (d) Sample size / respondent reach. When would you choose an Interview over a Questionnaire?

MODULE 4: ANSWER KEY & DETAILED RATIONALES

Question 1. B (Online questionnaire / survey)

Rationale: Surveys provide rapid, low-cost distribution across thousands of respondents, making them the only scalable option for 5,000 students.

Question 2. B (Passive Observation)

Rationale: Passive observation requires the analyst to watch silently without engaging or interrupting the subject.

Question 3. B (Hawthorne Effect)

Rationale: The Hawthorne Effect is the documented psychological reaction where subjects modify behaviors in response to being observed.

Question 4. B (Structured Interview)

Rationale: Structured interviews use a rigid, identical question sequence to eliminate interviewer variability across candidate responses.

Question 5. A (Document Analysis)

Rationale: Invoices, forms, and receipts provide exact historical data specifications without taking up any employee time.

Question 6. B (Semi-Structured)

Rationale: Semi-structured interviews allow an analyst to cover mandatory agenda items while freely investigating unexpected insights.

Question 7. B (Cannot clarify questions that respondents misunderstand)

Rationale: Once distributed, ambiguous questions cannot be re-explained in real-time, risking flawed survey responses.

Question 8. B (Active Observation)

Rationale: Active observation involves interrupting the user during their workflow to ask real-time clarifying questions.

Question 9. B (Documents often describe idealized formal policies rather than actual workarounds)

Rationale: Documents describe what should happen on paper; observation and interviews reveal what actually happens in reality.

Question 10. B (Closed-ended questions)

Rationale: Questions with constrained, predefined options (Yes/No, ratings, multiple choice) are closed-ended.

PART II: TECHNIQUE MATCHING ANSWERS

- 11. A (Interview) — Strategic executive direction requires one-on-one dialogue with top leadership.
- 12. B (Observation) — Staff rarely confess to breaking scanner protocols in interviews; observation catches the physical act.
- 13. D (Document Analysis) — Statutory tax rules and policies are documented in print with legal precision.
- 14. C (Questionnaire) — Reaching 10,000 mobile app users nationwide requires an automated digital survey.
- 15. A (Interview) — Diplomatic one-on-one interviews are needed to uncover reasons behind conflicting managerial viewpoints.

PART III: TRUE OR FALSE ANSWERS

- 16. FALSE — Closed-ended questions produce quantitative data that is very easy to analyze statistically.
- 17. TRUE — No single technique is sufficient; multi-method triangulations compensate for each technique's weaknesses.
- 18. FALSE — In active observation, the analyst specifically asks questions while observing the employee.
- 19. TRUE — Manual logs and manuals document official procedure, but employees frequently devise informal shortcuts.
- 20. FALSE — Requirements gathering is essential in every project to ensure the software solves actual user business needs.

PART IV: SCENARIO APPLICATION SAMPLE ANSWERS

Model Answer for Question 21 (First Technique for University Enrollment):

Recommended First Technique: Document Analysis.

Justification: An analyst should first collect and study all current enrollment forms, subject evaluation sheets, billing slips, and prerequisite guidelines. By doing this first:

- 1. The analyst learns the school's vocabulary, course codes, and database entities without wasting busy staff's time.
- 2. It prevents asking basic or repetitive questions during later interviews with the Registrar and Cashier.
- 3. It highlights confusing or outdated paperwork, allowing the analyst to prepare targeted, highly effective interview and observation questions.

Model Answer for Question 22 (Interviews vs. Questionnaires):

- **(a) Depth of Information:** Interviews provide rich, nuanced qualitative insights; questionnaires provide broad, shallow quantitative data.
- **(b) Cost & Time:** Interviews require high time commitment and financial expense per subject; questionnaires have extremely low marginal cost once prepared.
- **(c) Flexibility:** Interviews are highly adaptable (interviewer can probe unexpected answers); questionnaires are completely rigid.
- **(d) Sample Size:** Interviews typically involve 5–20 key stakeholders; questionnaires easily reach hundreds or thousands.

When to Choose Interviews: When dealing with high-level executives, exploring complex or political problems, or seeking detailed explanations of confusing business processes.